

Reinforcement Learning for American Puts Option

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Repository Link

<https://github.com/workkrishnpatel/IIT-B-ITSP-SOC-2026>

Story so far...

till week 7 we have seen the Neural Network training and that model performed so well . and then we have seen the theory and bit of practical exposure for the reinforcement learning with small Q table and now in week 8 we have trained the RL model for the policy wheather the stock must be exvcersised or kept on hold .

Why RL ?

Reinforcement learning is one of the three core pillars of machine learning . unlike supervised and unsupervised learning , we dont provide any targets or anything to the environment insted the RL works on the principle of the try and arrow methord . i mean that the agent learns by making mistakes and earns the rewards on being correct . at every wrong decision the agent got punishment of negati. be poiso since we we have already predicted the price through the neural network pricer now its time to look up into exercise or holding the stock whenever we 'll get the best output or best returns .

RL for American Puts Option

Environment :

The environment for this MDP is the GBM Geometric Brownian Motion . which stimulates the direction of the stock price and it gets stock prices over time by stimulating the finance models like GBM and volatility models over time .

State:

The state for this model is the Moneyness and time left to expiry the stock . that implies the moneyness is equal to the current stock price with ratio to the strike price and the time to expiry is the time period with ratio to the number of steps since this is discrete so that is the time period and by subtracting to this value by 1 its give the time left for expiry

Action:

As mentioned , this is the specific modeling to get the optimal stopping solution which gives the indication if the stock must be held or to exercise and also predict the exercise value

Reward:

This is the binary classification problem , so if the agent predict to hold then it will get 0 reward whereas on contrary the agent chooses to exercise immediately then as a reward it will get the $\max(K-S, 0)$ as . so the best and maximum reward we'll get that is the optimal path to consider

Component	Description
Environment	A simulated financial market where the stock price evolves according to Geometric Brownian Motion (GBM). The environment represents the American put option pricing problem and provides the agent with market information at every time step.
State	Each state consists of three normalized features: time fraction (t/T) , remaining time to maturity , and moneyness (S/K) . These features summarize the current market condition observed by the agent.
Action	The agent chooses one of two actions at every step: Hold (0) to continue the option, or Exercise (1) to immediately exercise the option and receive the intrinsic payoff.
Reward	If the option is exercised, the agent receives the discounted intrinsic value $\max(K - S, 0)$. If the option expires without exercise, the final

Component	Description
	discounted payoff is awarded. This reward structure encourages learning the optimal exercise timing.

Model Architecture

The modeling is done on Double Q Network . the architecture includes the following layers

- input layer consisting of three nodes the nodes depends on the number of state features since we have three features for this model i.e. moneyness , time period , time left for expiry
- 2 Hidden layers for this model with 64 nodes in both the layers and the activation function for this is ReLu .
- Output Layer consist of two nodes since this is the binary classification problem there are 2 Q value nodes.
- we have trained the model on the Adam Optimizer and with Huber loss function .
- Target network consist of the same or copy architecture of the above model
- we have used the gradient clipping , buffer replay . greedy policy for the good and optimize results

Hyperparameters and Configuration

Parameter	Value
Algorithm	Double Deep Q-Network (Double DQN)
Framework	TensorFlow / Keras
Training Episodes	10,000
Evaluation Episodes	5,000
Batch Size	128
Replay Buffer Size	50,000
Hidden Layers	2
Hidden Units	64
Learning Rate	0.0005

Parameter	Value
Discount Factor	Environment Discount Factor
Target Network Update	Every 250 Gradient Updates
Initial ϵ	1.0
Minimum ϵ	0.05
ϵ Decay	0.999

Results and Performance Summary

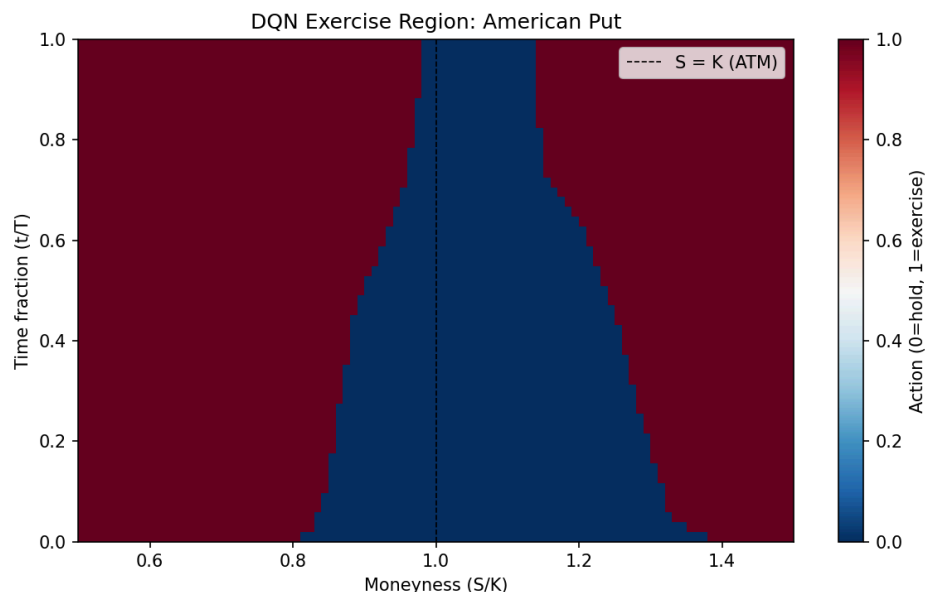
Policy	Option Value	Standard Error	Exercise Rate	Average Exercise Step
Always Hold	6.3631	0.1322	0.00%	—
Immediate Exercise	0.0000	0.0000	100.00%	0.0
Random Policy	0.7559	0.0241	100.00%	1.0
Double DQN (Proposed)	6.0866	0.0855	96.14%	28.0
CRR Binomial Benchmark	6.7233	—	—	—

The Deep Q Network agent successfully learns the policy by continuously interacting with the market stimulation GBM and gradually getting increment in reward with loss decreasing over 10000 episodes and also the cumulative reward is increasing steadily and with gradually improvement in the decision making that the rather than holding the exercise is more beneficial .

the exploration rate is decreased from 1 to 0.05 which implies that the agent explore initially and then it gets exploit in the conclusion phase . the uber loss was consistent throughout and also stabilize the convergence

The DQN achieve the american pricing of 6.0866 where as week 4 benchmark of CRR model estimated to 6.7233 which gives the absolute error of 0.6368 which indicates that the reinforcement learning agent successfully approximating the optimal solution of the option by while being closed to the analytical solution .

Exercise Boundary Plot



The exercise boundary plot provides a visual representation of the decisions learned by the Double DQN agent throughout the option's lifetime. The plot highlights regions where the agent chooses to exercise the American put option and regions where it prefers to continue holding the contract. The learned exercise region demonstrates that the agent has successfully identified meaningful decision patterns from the simulated market environment rather than making random choices.

The exercise boundary follows the expected behavior of an American put option, where exercise decisions become more frequent when exercising is economically advantageous. The boundary is reasonably structured and consistent, indicating that the neural network has learned a stable decision policy. While the learned boundary does not perfectly match the theoretical optimal exercise frontier obtained from the binomial model, it captures the overall characteristics of early exercise behavior and produces option values that remain close to the benchmark solution. This confirms that the reinforcement learning agent has learned a practical approximation of the optimal stopping strategy for American option pricing.

Conclusion

This project successfully demonstrates the application of Double Deep Q Networks for pricing American put options through reinforcement learning. By interacting with a simulated financial market environment, the agent learned when to exercise the option and when to continue holding it in order to maximize the expected discounted payoff.

The final trained model achieved an option value of 6.0866 compared to the Cox Ross Rubinstein benchmark value of 6.7233, indicating that the learned policy closely approximates the theoretical solution. Stable training behavior, consistent reward improvement, and strong performance on unseen evaluation paths show that the agent was able to learn an effective exercise strategy from experience.

Overall, the project highlights the effectiveness of deep reinforcement learning for solving optimal stopping problems in quantitative finance and demonstrates how neural network based agents can learn complex financial decision making without relying on explicit analytical pricing formulas during training.

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